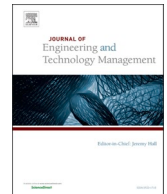




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The moderating effect of absorptive capacity on the technology search and innovation quality relationship in high-tech manufacturing firms

Yunlong Duan^a, Lei Huang^{b,*}, Xuan Luo^a, T.C.E. Cheng^c, Hanxiao Liu^a^a International Business School, Yunnan University of Finance and Economics, Kunming, China^b School of Management, Zhejiang University of Technology, Hangzhou, China^c Department of Logistics and Maritime Studies, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

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ABSTRACT

Based on the knowledge-based view, we conceptualize the technology search of high-tech manufacturing firms into three dimensions, namely search breadth, search depth, and balanced search, and construct the functional paths of the relationships between the different technology search dimensions and innovation quality of high-tech manufacturing firms. Grounded in absorptive capacity theory, we analyze how such relationships will change with the absorptive capacity level. Using data from China's high-tech manufacturing industry from 2009 to 2017, we construct a panel data model to test the moderating effect of absorptive capacity on such relationships. Our research findings suggest: (i) Technology search breadth and depth each have a significant, inverted U-shaped relationship with innovation quality, i.e., technology search breadth and depth have an optimal point, and innovation quality exhibits an initial increasing and then decreasing trend. (ii) Balanced search has a significant positive relationship with innovation quality, i.e., there is a balanced point between technology search breadth and depth, and the higher the balance point is, the higher is innovation quality. (iii) Absorptive capacity does not merely positively moderate the relationship between technology search and innovation quality, but has different moderating effects under different search levels.

1. Introduction

In the high-tech industry, where technology is rapidly developing, high-tech manufacturing firms can only meet the ever-changing market demand by continuously innovating and constantly improving the quality of innovation and by enhancing market competitiveness to be formidable in competition. Currently, facing a dynamic external competitive environment, changing technological knowledge of firms, and problems such as the “capability trap” and “innovator dilemma” (Duan et al., 2021; Nerkar, 2003; Swift, 2016), many high-tech firms have embarked on the journey of open innovation. In the context of open innovation and rapidly changing business environments, “closed-door” innovations cannot cope with the era of accelerating product renewal, and firms have to conduct technology search at various stages of innovation (Dong and Netten, 2017; Escribano et al., 2009; Hameed et al., 2012). Through

* Corresponding author.

E-mail addresses: duanyl1977@126.com (Y. Duan), Hwang0813@qq.com (L. Huang), luoxuaning@126.com (X. Luo), edwin.cheng@polyu.edu.hk (T.C.E. Cheng), 932131741@qq.com (H. Liu).

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technology search and acquiring and integrating external technical knowledge, high-tech manufacturing firms can expand and update their knowledge bases, overcome the limitations of internal technical knowledge, track advanced technologies, and better deal with problems such as high cost and complexity in the process of technological development (Enkel et al., 2009; Huizingh, 2011; Tan et al., 2020). Spithoven et al. (2010) proposed from the perspective of innovation input that in open innovation, firms do not rely entirely on internal R&D but scan the environment to search for technology and knowledge. In the increasingly open and rapidly changing industrial environment, manufacturing firms need cross-integration of heterogeneous knowledge from different fields and industries to generate new ideas and technologies through “creative collision”, so external technology search has become an important source and channel for firm innovation resources (Laursen and Salter, 2014).

In recent years, there is the phenomenon of “quantity over quality” in Chinese firms’ pursuit of innovation. According to national statistics, China has the world’s second largest expenditures on R&D, behind the US in total expenditures and exceeding the average level of the 15 countries in the EU. Specifically, China ranks first in the world in total R&D personnel, and in invention patent applications and authorizations. The number of scientific and technological papers published in China accounted for 26.9 % of the global total published in 2015–2017, followed by the US with 16.8 %.¹ Although the number of innovative achievements of China is in the world’s leading position, the quality and effectiveness of technological innovations are still far from those of the major developed countries. The fundamental reason is that the state and firms have been focusing less on innovation quality and more on innovation quantity. Since the financial crisis in 2008, and with the experience of the recent Sino-US trade war and ZTE incident, Chinese manufacturing firms, especially high-tech manufacturing firms, have undergone major transformations in both material and spiritual aspects. Innovation quality is the sum of innovation performance in processes, outputs, and economic benefits (Haner, 2002). Currently, most of China’s manufacturing industries are largely at the bottom of the industrial value chain due to the lack of core technologies (Yuan and Li, 2021), e.g. semiconductor, pharmaceutical, electronics, frontier equipment and other manufacturing industries. Low-quality innovation leads to low economic performance. Only high-quality innovation with heterogeneity can yield competitive advantage for firms and improve their performance (Hirshleifer et al., 2013; Tan et al., 2020). Existing empirical studies have confirmed the important role of innovation quality in firm development.

A firm’s innovation quality lies in the industrialization of innovative achievements, e.g., high level patents, scientific and technological breakthroughs, and R&D of new products, which generates profits and lead to continuous improvement of core competency, not just the number of patent authorizations, number of scientific and technological papers published, or number of new-product developments. According to the 2018 *China Patent Investigation Report* released by the State Intellectual Property Office in January 2019, the industrialization rate of effective invention patents in 2018 was only 32.3 %, down from 36.7 % in 2016 and 36.2 % in 2017. Therefore, firms in the development process, especially high-tech manufacturing firms, urgently need to improve the efficiency and quality of their innovation efforts after investing large amounts of R&D resources. However, the quality of innovation is inevitably influenced by the new technical knowledge that R&D personnel have searched before proposing ideas. If the technology used to develop a product is outdated or is not the future development direction of the market, it is difficult to realize the profit and commercialization of the new product in the market, even if there are many patent applications. Through conscientious searches for heterogeneous knowledge elements, high-tech firms realize the rapid transfer of high-quality information and tacit knowledge (Duan et al., 2021; Gay, 2014), integrate advanced technical knowledge into their R&D processes, create new combinations of technology and knowledge, and achieve integrated innovation. Then, the new products developed tend to be more easily recognized by the market and customers, and the innovation efficiency and core competency of the firms will also be improved (Bolívar-Ramos et al., 2012; Leiponen and Helfat, 2010; Tan et al., 2020), thereby improving their innovation quality.

The improvement of innovation quality of high-tech manufacturing firms cannot be separable from their searches for technical knowledge, whether the goal of high-quality innovation can be achieved after acquiring the necessary technical knowledge is closely related to their absorptive capacity (Tamayo-Torres et al., 2011; Wang et al., 2020). The new technological knowledge acquired by firms through technology search can produce innovative output through a series of internal processes in which absorptive capacity plays an important role in helping them to generate high-quality innovation output (Fosfuri and Tribó, 2008; Petti et al., 2019). Absorptive capacity is the dynamic ability of the innovation subject to identify, acquire, digest and utilize valuable external knowledge and information through a series of learning processes (Lane et al., 2006; Martinkenaite and Breunig, 2016; Zahra and George, 2002). The reason is that absorptive capacity can reduce the potential threats and risks in the process of technology search, and the firm that has high absorptive capacity can identify high-quality knowledge elements beyond the technological boundary, bringing high-quality innovative output to the firm (Huang and Rice, 2009; Mir et al., 2016). Researchers have criticized that existing research places too great an emphasis on the impact of external contact on firm innovation, ignoring the series of processes through which firms digest and transform knowledge. Differences in firms’ ability to absorb external technical knowledge lead to uneven innovation performance (Cohen and Caner, 2016; Giuliani and Bell, 2005), with firms having low absorptive capacity stagnating at low levels of innovation quality.

In view of the globalization of competition, and an increasingly complex and dynamic market environment, how to improve the innovation quality of China’s high-tech manufacturing firms and maintain their core competency are critical management issues to address. In this study, seeking to deepen and enrich search and innovation theories, we use data from China’s high-tech manufacturing industry as a sample to explore the relationship between the technology search behaviour of high-tech manufacturing firms and their quality of innovation. Grounded in absorptive capacity theory, we explore the moderating effect of absorptive capacity to expand the

¹ The data come from the *Statistical Bulletin of National Economic and Social Development* in 2018.

contingent factors that affect innovation quality. Our findings provide useful guidance to high-tech manufacturing firms to improve their innovation quality and achieve industrial transformation and upgrading.

2. Literature review and research hypotheses

2.1. Related concepts and dimensions

2.1.1. Technology search

The concept of technology search stems from the organization search theory proposed by Winter and Nelson (1982). Researchers hold different views of technical search based on different perspectives. However, there is agreement among the different views that technology search is a behavioral process based on a technical perspective to solve technical problems or find new opportunities for an uncertain future by sourcing innovative knowledge inside and outside an organization to achieve innovation goals. Examples include Cruz-González et al. (2015); Lichtenthaler (2011); Rohrbeck (2010); Tan et al. (2020) etc.

Existing research on the dimensions of technology search is based on different perspectives, including such single dimensions as the time dimension (Nerkar, 2003), spatial dimension (Ahuja and Katila, 2004), organizational dimension (Miller et al., 2007), and cognitive dimension (Rosenkopf and Almeida, 2003). Researchers have also discussed the synergistic effect of two-dimensional technology search, e.g., Katila (2002) and Phene et al. (2006). In addition, based on the extent to which firms use existing technical knowledge and the scope of technical knowledge sources, Guan and Liu (2016); Li et al. (2008) and Katila and Ahuja (2002) classified technology search into search depth and search breadth, which have been widely recognized in academia.

On the basis of existing research, combined with the current innovation status of high-tech manufacturing firms in China and with reference to Ahuja and Katila (2004); Cao et al. (2009); Laursen and Salter (2006) and Guan and Liu (2016), we conceptualize technology search in the context of high-tech manufacturing into three dimensions, namely search breadth, search depth, and balanced search (see Fig. 1). Technology search width is the scope of searching for new technology knowledge, technology search depth is the extent to which firms reuse existing technology knowledge, and balanced technology search means that firms maintain a relative balance between search depth and search width in the process of technology search (Ahuja and Katila, 2001; Cao et al., 2009; Guan and Liu, 2016; Laursen and Salter, 2006). The reasons for such a classification system are as follows: First, the breadth and depth dimensions of technology search are helpful for high-tech manufacturing firms to formulate an effective search strategy for technical knowledge under the constraint of limited organizational resources such that they have a definite target, and can improve their innovation efficiency and quality. Second, under the constraint of limited resources, expanding the breadth of technology search is conducive to increasing firms' heterogeneous knowledge and enriching their knowledge bases. In addition, absorbing and digesting the heterogeneous knowledge is conducive to eliminating firms' knowledge blind spots, enabling them to better cope with environmental turbulence (Laursen and Salter, 2006). Third, increasing the depth of technology search can help firms gain a deeper understanding of some knowledge so that they can leverage the heterogeneous knowledge to improve their innovation ability (Laursen and Salter, 2006). Fourth, in practice, a single search strategy will not be able to adapt to the rapidly changing market environment and shifting strategic focus, while multiple search strategies will inevitably give rise to resource competition under the constraint of limited firm resources, thus affecting the efficiency and quality of innovation (Gatti et al., 2015). There is a tension between search breadth and depth under the constraint of organizational resources and self-reinforcing mechanisms, which drives the search to an unbalanced state (Gupta et al., 2006), while firms' innovation focusing on search balance is more robust (Lee, 2015).

2.1.2. Absorptive capacity

Cohen and Levinthal (1990) first proposed the concept of absorptive capacity, which states that whether a firm can effectively

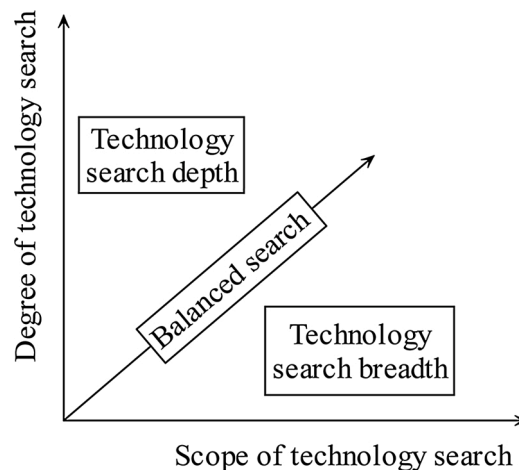


Fig. 1. Technology search dimensions.

screen, evaluate, and utilize external knowledge depends on the level of the firm's absorptive capacity, which is its ability to identify, digest, and use the value of external new information for commercial purposes. Later, researchers, based on dynamic capability theory and the process perspective, supplemented and complemented the discussion (e.g. Lane et al., 2006; Zahra and George, 2002; Gkypali et al., 2018).

Although there are different classifications of absorptive capacity in the literature, they are mainly based on the three-dimensional model comprising the acquisition, assimilation, and application of external knowledge proposed by Cohen and Levinthal (1990), and Lewin et al. (2011), and on the four-dimensional model consisting of acquisition, assimilation, transformation, and application of external knowledge proposed by Zahra and George (2002). In most of the existing empirical studies on absorptive capacity, researchers generally agree with the four-dimensional model of Zahra and George (2002).

At present, research on absorptive capacity is well developed, and researchers have come to a consensus that absorptive capacity has process attributes and dynamic ability attributes. In general, the absorptive capacity of a firm is the process of organizational learning and knowledge management, including identification and acquisition of technical knowledge, application and integration of knowledge elements, and commercialization of new technical knowledge.

2.1.3. Innovation quality

Prior studies on innovation quality are mainly focus on the innovation and quality perspectives (Haner, 2002; Lahiri, 2010; Mu et al., 2017; Wang et al., 2018; Duan et al., 2021). The concept is developed on the basis of innovation theory and quality management theory. Based on innovation theory, research from the perspective of innovation holds that innovation quality is the degree of influence of innovation output, embracing innovation ability and innovation performance. Haner (2002) first proposed the conceptual framework of innovation quality, arguing that innovation quality is the sum of innovation performance in various fields and has special dynamics that can be defined by variables such as quantity, performance, efficiency, reliability, time, cost, and value to customers. Lahiri (2010) argued that innovation quality is the impact of innovation output and an important embodiment of innovation ability. Researchers have advocated that innovation quality be measured by using the authorization rate, number of times cited, and length of a patent. Based on quality management theory, innovation quality is regarded as the embodiment of the innovation process quality and the degree to which innovation results meet the market demand, which is different from quality innovation. Haner (2002) proposed that innovation quality is the quality of all the innovation results, including product or service quality, process quality, and firm quality.

High-tech manufacturing firms produce high-tech products (Raab and Kotamraju, 2006), of which knowledge innovation and product innovation are their most prominent features. Based on the studies of Haner (2002), Mu et al., 2017; Palm et al., 2016; Wang et al., 2018; Tseng and Wu, 2007) etc., We regard innovation quality as the adaptability of the market and the satisfaction of the interests of all parties to the results of innovative products such as new products or services, new processes or technologies, and new management models created by firms using various innovative resources, as a comprehensive manifestation of the ability, performance, efficiency, and value of corporate innovation activities in terms of process and results, as a standard for measuring and judging the level and quality of firm innovation output.

2.2. Technology search and innovation quality in high-tech manufacturing firms

Current academic research on technology search and innovation mainly discusses the relationship between technology search, and innovation capabilities and innovation performance. Some scholars, such as Leiponen and Helfat (2010), and Phelps (2010), have found that external technology search has positive effects on firms' innovation capabilities and innovation performance. The continuous enhancement of search contributes to the rapid transfer of high-quality information and tacit knowledge, and improves the efficiency of firm innovation (Gay, 2014). Some scholars have also found that technology search has a negative effect on innovation performance, e.g., Knudsen and Mortensen (2011) proposed that increasing the openness of external knowledge sources at the project level would have a negative impact on new product innovation performance. Although firms can search for technical knowledge in depth, they can easily fall into "core rigidity" and a "capability trap", hindering and inhibiting firm innovation (Swift, 2016). With the deepening of research, scholars have found that the relationship between technology search and innovation performance is not a simple linear relationship. For example, (Laursen and Salter, 2006), and Wu (2013) took British and Taiwanese firms in China as samples, respectively, and detected an inverted U-shaped relationship between technology search and innovation performance.

To summarize, there are many research achievements relating to firm technology search in the field of innovation, but the research still needs to be further deepened. First, existing studies focus on the relationship between technology search, and innovation performance and innovation capabilities. Few studies take innovation quality as a dependent variable to explore the relationship between technology search and innovation quality. Second, there are still differences in existing findings, mainly with regard to the linear or nonlinear relationship between technology search, and innovation performance and innovation capabilities. Third, in terms of the impact of different search dimensions on innovation performance, although scholars have abandoned the single dimension in recent years and studied the impact of multi-dimensional search on innovation, few have considered the impact of multi-dimensional balanced search on innovation. In addition, most existing studies only focus on the direct impact of technology search on innovation, paying scant attention to the contingency factors that affect the relationship between technology search and innovation.

At present, in the context of China's economic and social transition, searching for and utilizing external technical knowledge to improve the efficiency and quality of technological innovation is an inevitable choice for Chinese firms to integrate into an open innovation system (Chen et al., 2011; Duan et al., 2021). Innovation quality is an important index to measure the level and height of innovation, and the integration of innovation performance in each field. Innovation quality and innovation performance are

interrelated. Therefore, based on the relationship between innovation quality and innovation performance, it is necessary to analyze the relationship between technology search and innovation quality. In view of past studies, we divide the technology search dimension into search breadth, depth and balanced search, and then analyze the relationship between different types of search and innovation quality.

2.2.1. Search breadth and innovation quality in high-tech manufacturing firms

Knowledge sources outside the firm play an important role in acquiring new knowledge, inspiring new thinking, and generating new ideas; technology search describes the scope to which a firm searches for new technology knowledge. Different types of knowledge provide firms with a more diversified knowledge base, which is beneficial for firms to carry out exploratory innovation activities through a unique combination of knowledge (Choi et al., 2018; Subramaniam and Youndt, 2005). First, increasing the breadth of search is conducive to expanding the sources and scope of external innovation resources acquired by firms and will increase the heterogeneous knowledge acquired by firms from external search. It is helpful for firms to increase opportunities for the reorganization and application of existing technologies, and generate diversified selection effects (Ahuja and Katila, 2004). Firms can effectively reduce the risk and uncertainty of innovation, improve the probability of innovation success and promote the improvement of innovation quality by reorganizing, absorbing and utilizing the heterogeneous knowledge obtained from search (Cohen and Levinthal, 1990; Fleming and Sorenson, 2004). Second, the firm's extensive technology search can select valuable high-quality innovation elements from many heterogeneous resources, effectively breaking the resource bottleneck of high-tech manufacturing firms (Sampson, 2007). The combination of these key heterogeneous resources will also bring new ideas to firms, increase the potential for firms to explore new knowledge and novel solutions (Chiang and Hung, 2010; Choi et al., 2018), and generate more high-quality innovative outcomes. Finally, abundant heterogeneous resources can not only meet the demand of innovation but also pose threats and challenges to the firms' current cognitive paradigm, thereby increasing the possibility of cooperation between firms and external heterogeneous knowledge sources, promoting firms to search, absorb and utilize knowledge, improving organizational learning efficiency and stimulating the improvement of the quality of innovation (Fleming and Sorenson, 2004).

However, the breadth of firm technology search is not as wide as possible. With the increase of search breadth, the financial and cognitive costs of search will also increase, thus decreasing the marginal benefit to firms from integrating and utilizing knowledge (Fleming and Sorenson, 2004). Firms have limited ability to learn and utilize the new technical knowledge they have found, which will increase the cost of knowledge reconstruction and may lead to knowledge redundancy (González-Moreno et al., 2019; Wu, 2013). In particular, the increasing complexity of SMEs' extensive search for diverse knowledge will bring significant challenge to the innovation activities of firms (Leiponen and Helfat, 2010). Second, the advanced technical knowledge found only represents the potential for future innovation; its implementation requires much investment (Kotha et al., 2013). In a high-tech industry with rapid technological upgrading, innovation risks are relatively high. Due to the limitations of scale and financial strength, excessive search may cause firms to take higher risks and squeeze innovation resources, which is not conducive to firm innovation.

H1. Technology search breadth has an inverse U-shaped relationship with innovation quality in high-tech manufacturing firms. In other words, with an increase in technology search breadth, the quality of firm innovation is on the rise; when technology search breadth exceeds the inflection point level, the quality of firm innovation begins to decline.

2.2.2. Search depth and innovation quality in high-tech manufacturing firms

In the process of technology search, firms, especially SMEs, are more inclined to increase their search depth due to limitations on their own resources (Guo et al., 2015). Search depth reflects the extent to which a firm reuses existing knowledge elements to identify new opportunities. First, the enhancement of search depth helps maintain the relevance of external sources of knowledge to the firm (Lavie, 2007). Searching in relatively familiar fields is convenient to form and accumulate valuable tacit knowledge, summarize its utilization rules and improve its potential defects, produce newly technical knowledge that is difficult to imitate, form the technical barriers of the firm itself, and promote innovation quality. With the continuous deepening of search, firms' knowledge discrimination ability is enhanced, utilization of erroneous knowledge resources is avoided, the reliability of external knowledge search is improved, and the innovation efficiency and success rate of firms are improved (Gay, 2014). Second, in-depth search in familiar fields reduces the diversity of knowledge, reduces the complexity of management, and helps firms to quickly match the external knowledge they find with their own knowledge and reduce the risk and cost of innovation (Kaplan and Vakili, 2015). In-depth search of existing knowledge can effectively promote communication and interaction between firms and the outside world and knowledge sharing, optimize the allocation of innovation resources among subjects to produce a "synergy effect", and gradually establish cooperative practices and tacit understanding, thus leading to higher quality innovation outcomes (Dhanaraj et al., 2004).

Despite the reliability and cost advantages of deep search, search depth may not be as deep as possible based on the entire innovation process. In a familiar field, the knowledge obtained by deep search lacks inspiration and diversity, and a single knowledge situation will generally reduce the sensitivity of firms to changes in the external environment (Duan et al., 2021; Laursen and Salter, 2014). It is not possible to quickly perceive customer preferences, and it is difficult to track the frontier of technology in time; firms are thus unable to effectively cope with dynamic competition, form a mindset and overcome organizational inertia, which causes firms to experience a "lock-in" effect, fall into the capability trap, and transform core competitiveness into core rigidity (Swift, 2016), thus hindering the improvement of firm innovation quality. Second, from the law of diminishing marginal utility, it can be seen that firms continue to strengthen the depth of utilizing the same knowledge, and their incremental innovation revenue is decreasing (Choi et al., 2018; Fu, 2012). Based on the above analysis, this paper proposes the following research hypothesis:

H2. Technology search depth has an inverse U-shaped relationship with innovation quality in high-tech manufacturing firms. In

other words, with an increase in technology search depth, the quality of firm innovation is on the rise; when technology search depth exceeds the inflection point level, the quality of firm innovation begins to decline.

2.2.3. *Balanced search and innovation quality in high-tech manufacturing firms*

Due to the influence of organizational resources and organizational culture, there is a certain tension and contradiction between the depth and breadth of technology search (March, 1991). The breadth and depth of technology search make competitive demands on organizational resources, and firms are faced with the problem of resource allocation and balance when choosing a search strategy. In a dynamic and complex market environment, firms not only need to expand their search breadth to acquire diversified and heterogeneous technical knowledge to adapt to the changing environment but also need to deepen the search depth of existing technical knowledge to maintain competitive advantage and market share. Due to uncertainty of the environment and limited organizational resources, excessive emphasis on search breadth, although it can update the knowledge base of the firm, will lead to the results of “suboptimal balance” and “path dependence”; excessive emphasis on search depth can dig deep into existing technologies, but it cannot quickly adapt to changes in the environment and is prone to fall into “core rigidity” and “capability trap”, thus hindering and inhibiting firm innovation (Swift, 2016). Therefore, how to coordinate the tension and contradiction between the breadth and depth of technology search is a problem that cannot be ignored when high-tech manufacturing firms implement a search strategy.

This paper refers to the studies of Cao et al. (2009) and Gupta et al. (2006) to introduce the balanced search dimension. Balanced search is a relatively balanced state between search breadth and depth intended to reduce search risks and maximize search benefits when a firm conducts the strategic behavior of technology search. On the one hand, in-depth search is carried out in familiar fields, which allows firms to explore some specialized knowledge features and knowledge correlation rules, facilitate a deep understanding of relevant knowledge, and develop combination products with novel technologies and functions to meet market demands to improve the efficiency of innovation. However, by giving too much attention to search depth and ignoring its breadth, it is easy to form a certain cognitive inertia and let firms tilt toward the familiar environment (Beckman et al., 2004), thereby limiting firms' ability to absorb external knowledge and hindering their improvement of innovation quality. It is also difficult to adapt to the presently popular “crossover” thinking, which will expose firms to the risk of knowledge obsolescence. On the other hand, if a firm focuses too much attention on search breadth but ignores search depth, it will bear a higher cost and risk. After all, not all the knowledge found can be immediately applied to current innovation activities, so it is easy to experience difficulty in obtaining timely returns. Therefore, to avoid the above extreme situation, firms need to focus on the duality of technology search, especially high-tech manufacturing firms with high investment in research and development and high risk, and they need more balanced support for both to reduce search risk and maximize search benefit. Based on these needs, this paper proposes the following research hypothesis:

H3. Balanced search positively impacts on innovation quality of high-tech manufacturing firms. In other words, the more the breadth and depth of technology search tend to be balanced, the better the firm innovation quality is.

2.3. *The moderating effect of absorptive capacity*

Absorptive capacity is the ability to internalize and apply external knowledge; it is an indispensable contingency factor in studying the relationship between external technology search and innovation. Current considerations for innovation contingency factors include existing research mostly on external factors, such as environmental turbulence and external networks, and also include the internal factors of the firm, such as internal resource constraints and the knowledge base of the firm itself. However, the firm's ability to absorb the acquired technical knowledge after the search is not enough. In fact, for high-tech manufacturing firms, the goal of improving innovation ability and innovation quality can only be achieved by fully integrating and applying diversified new knowledge elements acquired through technology search into internal knowledge under the role of their own absorptive capacity.

The existing research results show that there are three main paths for absorptive capacity to influence firm innovation. First, it directly influences the firm innovation (Kostopoulos et al., 2011). Second, it has a mediating effect on firm innovation (Gkypali et al., 2018; Lavie et al., 2010). Third, it has a moderating effect on firm innovation (Patel et al., 2015; Zobel, 2017). Based on the path constructed in the previous section for relationships between different technology search dimensions and the innovation quality of high-tech manufacturing firms, the absorptive capacity theory is introduced to analyze how the relationship between the technology search and innovation quality of high-tech manufacturing firms will change under different absorptive capacity levels. The reasons lie in the following two aspects:

First, existing studies on contingency factors of firm innovation mainly focus on external factors of firms, and there are few studies on the relationship between firms' own absorptive capacity and innovation. However, the innovation quality of firms have a great relationship with their own absorptive capacity.

Second, current research on the moderating effect of absorptive capacity, although there are many research results, mainly discusses its moderating effect on firm innovation performance (Wang and Han, 2011) and new product innovation (Zobel, 2017); few studies have involved the moderating effect on firm innovation quality. Firms conduct technology search behaviors for innovation and on whether the quality of their innovation results will be moderated by the firm's own absorptive capacity (Petti et al., 2019). There are no studies on this topic at present. Therefore, for this paper, after analyzing the relationship between different dimensions of technology search and innovation quality of high-tech manufacturing firms, it is necessary to introduce absorptive capacity to analyze its moderating effect.

The key to achieving improvement in high-tech manufacturing firms innovation quality is not only to acquire heterogeneous knowledge from external sources through technology search but also to have the ability to integrate and apply such knowledge.

Technology search provides potential possibilities to create new knowledge and products for a firm. How to internalize the searched external knowledge into their own technology, innovate and form a core competitive advantage, and develop absorptive capacity is a key factor (Enkel and Gassmann, 2010; Lu et al., 2016; Petti et al., 2019). Rothaermel and Alexandre (2009) found that the differences in firm's absorptive capacity lead to significant differences in the breadth and depth of external technology search.

The strategic behavior of increasing firm search breadth is to expand the scope of technology search and search for heterogeneous knowledge elements from suppliers, customers, competitors, partners, intermediaries, universities and scientific research institutions. However, what is searched with a wider scope is a diversified new knowledge element that has relevance differing from that of the firm's previous knowledge. How to transform diverse new knowledge elements into ones that can be used by the firm and applied to the development of products, services or technologies is the key to the success and efficiency of firm innovation (Leal-Rodríguez et al., 2014; Wang et al., 2020). Firms with high absorptive capacity can effectively manage diversified and heterogeneous knowledge, identify the "blank space" between low-correlation knowledge, explore innovation opportunities, and improve level of innovation quality. Existing studies have shown that the innovative output of firms that continue to invest in their absorptive capacity is more consistent with the demands of different markets (Gkypali et al., 2018; Wang et al., 2020). Thus, the stronger the absorptive capacity of firms, the greater is the degree of diversified heterogeneity knowledge marketization and the greater the quality of their innovation output is. Therefore, this paper proposes the following research hypothesis:

H4a. Absorptive capacity positively moderates the relationship between search breadth and innovation quality. In other words, the inflection point of the inverted U-shaped curve relating technology search breadth with innovation quality in high-tech manufacturing firms with high absorptive capacity moves to the right compared with that of firms with low absorptive capacity.

The strategic behavior of firms to strengthen their search depth is to conduct a deeper search for existing technical knowledge to achieve technical renewal and to upgrade products and services through knowledge reorganization (Enkel and Heil, 2014; Petti et al., 2019; Wang et al., 2020). In-depth technology search strengthens firm's understanding of existing knowledge, improves the utilization efficiency of similar knowledge elements, and reduces the cost and risk of searching. A firm with low absorptive capacity has a low degree of in-depth integration and effective transformation of the searched knowledge elements, and the efficiency and quality of innovation are low. A firm with high absorptive capacity can find opportunities for the integration of heterogeneous knowledge elements and realize synergies of knowledge to carry out product improvement and new product development in a more targeted way. At this time, strengthening search depth can promote the transformation of high value and high-quality knowledge and promote the integration of new knowledge and innovation activities to improve the innovation quality. Thus, firms with strong absorptive capacity are better at reorganizing and changing old knowledge, which can better promote the relationship between search depth and innovation quality. Therefore, this paper proposes the following research hypothesis:

H4b. Absorptive capacity positively moderates the relationship between search depth and innovation quality. In other words, the inflection point of the inverted U-shaped curve relating technology search depth to innovation quality in high-tech manufacturing firms with high absorptive capacity moves to the right compared with that of firms with low absorptive capacity.

Balanced search is also a strategic behavior of technology search, which aims to maintain a relative balance between search depth and breadth. In the process of firm technology search, increasing the search distance along any dimension may bring both positive and negative effects. The moderating effect of absorptive capacity is as follows: First, the identification and acquisition ability of absorptive capacity can not only increase the breadth and depth of technology search and strengthen the potential connection among knowledge but can also shorten the development cycle of new products and accelerate the speed of new products entering the market, thus improving the quality of innovation (Petti et al., 2019; Vasudeva and Anand, 2011). Second, the digestion ability of absorptive capacity can connect new technical knowledge with owned technical knowledge and accelerate the speed of high-quality innovation outcomes. Moreover, digesting external knowledge can avoid repeated searching. Third, the transformation ability of absorptive capacity can

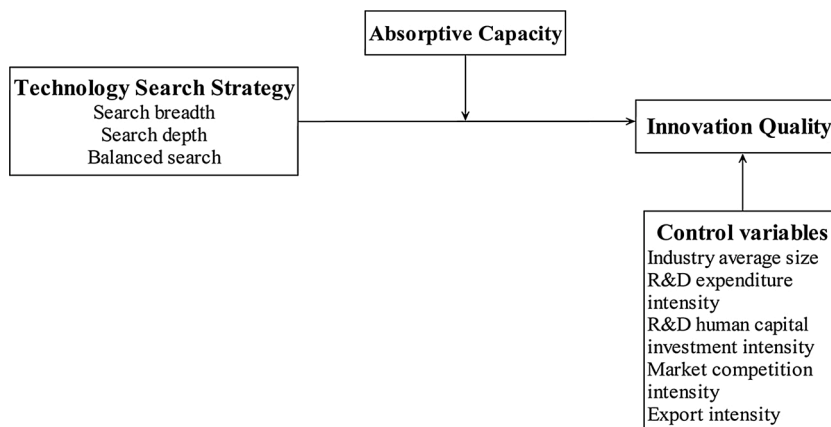


Fig. 2. Research model.

integrate the heterogeneous knowledge searched by firms into new knowledge, avoiding the path dependence of knowledge (Duan et al., 2021; Yu et al., 2019). In the process of transformation, new ideas can be generated to reconstruct the firm's cognition of the new external situation and identify new opportunities. Fourth, whether the search work of firms needs to be broadened and deepened depends on whether the search knowledge can meet the needs of firms to solve problems. The application ability of absorptive capacity can incorporate transformed new knowledge into the practical innovation of the firm (Duan et al., 2021; Wang et al., 2020). Therefore, this paper proposes the following research hypothesis:

H4c. Absorptive capacity positively moderates the relationship between balanced search and innovation quality. In other words, compared with firms with low absorptive capacity, the slope of the curve relating balanced search and innovation quality in high-tech manufacturing firms with high absorptive capacity will change.

Based on the above analysis, the research model of this paper is shown in (Fig. 2).

3. Methods

3.1. Data

The source of data for this study is from the China Statistical Yearbook on China's High Technology Industry, which has been collected and published by the National Bureau of Statistics. The Yearbook contains the production and operation status, research and development activities, new product development and sales, and patents of all high-tech manufacturing firms in China. The Statistical data of all high-tech firms are classified by industry sector, region (province), ownership, and scale of firms. This paper selects industry data as a research sample. The reasons are as follows: First, The Yearbook includes all high-tech firms in China. This paper selects the data of high-tech manufacturing firms summarized by industry sector in the Yearbook to explore the relationship between technology search behavior and innovation quality of firms of the same type—all firms belonging to the same industry (Rosenkopf and Nerkar, 2001; Tan et al., 2020; Yu et al., 2019). Second, it is the only way we can obtain comprehensive and complete high-tech firm data, and the data is true and reliable due to the legal obligation of true reporting.

According to the current industrial classification of China's national economy, high-tech manufacturing mainly includes six major categories, namely pharmaceutical manufacturing, aerospace and equipment manufacturing, electronic and communication equipment manufacturing, computer and office equipment manufacturing, medical equipment and instrumentation manufacturing and information chemical manufacturing. There is a lack of data on R&D activities in China Statistical Yearbook on China's High Technology Industry before 2009, and the information chemicals manufacturing industry only has data for 2015 and 2016. To fit the data availability to the balanced panel, this paper selects the panel data of five high-tech manufacturing industries from 2009 to 2017 for empirical analysis. (Considering the hysteresis of the influence of variables, the dependent variable innovation quality is the data from 2010 to 2017, while the other variables are the data from 2009 to 2016.) We use STATA 15.1 to process and analyze the sample data.

3.2. Measures

3.2.1. Dependent variable

Existing studies mainly measure the quality of innovation by adopting the first four places of patent IPC classification number, patent citation rate and citation times, patent authorization rate and patent length, etc. (Bloom and Van, 2002; Lahiri, 2010; Lerner, 1994). Comino and Graziano (2015) holds that innovation quality is an important index to measure innovation level and height, and the number of patents will affect innovation quality. The types of patents in China are divided into invention patents, utility model patents and design patents. In particular, invention patents refer to new technical solutions for products, methods or improvements, which are regarded as representative of high-quality innovation output in China (Duan et al., 2021; Yu et al., 2019). Based on the availability of data and the reliability of research results, this paper refers to the study of Duan et al. (2021) and Yu et al. (2019), and the innovation quality is measured by the ratio of the granted number of invention patents to the number of patent applications.

3.2.2. Independent variables

Current studies mostly use patent citation data and questionnaires to measure technical search variables, but patent citation in developing countries does not objectively reflect the level and direction of search (Laursen and Salter, 2006). Based on the availability of data and the reliability of research results, this paper selects high-tech manufacturing data from official statistics to measure technology search. There is an indicator of technology expenditure in China Statistical Yearbook on China's High Technology Industry. Technology expenditure includes four sub-indicators, namely technology introduction expenditure (TIE), purchase domestic technology expenditure (PDTE), technology digestion expense (TDE), and technological transformation expenditure (TTE). Total technology expenditure is the sum of these four sub-indicators.

The technology search breadth (Sb) reflects the extent of searching new knowledge or technology and the channels of its source. The breadth of search can be replaced by "search scope" (Choi et al., 2018; Katila and Ahuja, 2002). In terms of geospatial dimension, the scope of new knowledge and new technology sources is divided into local (domestic) and non-local (foreign) (Laursen, 2012)—that is, the purchase of new technical knowledge from within the country or the introduction of new technical knowledge from abroad. Therefore, this paper takes the proportion of the sum of purchasing domestic technology and importing technology from abroad costs accounting for total technical expenditure as the proxy variable to measure the technology search breadth of high-tech manufacturing firms. Its calculation formula is $Sb = (TIE + PDTE) / \text{Total technology expenditure}$.

The technology search depth (Sd) reflects the degree of integrating and utilizing the existing knowledge. Laursen and Salter (2006) believe that after the absorption and digestion of searched heterogeneous technical knowledge, a deeper understanding of information correlations will result: the greater the degrees of digestion, absorption and transformation are, the more flexible their application will be, and the higher the efficiency of problem solving will be (Cruz-González et al., 2015). Therefore, this paper takes the proportion of the sum of digesting and absorbing technical knowledge costs accounting for total technical expenditure as the proxy variable to measure the technology search breadth of high-tech manufacturing firm. Its calculation formula is $Sd = (TDE + TTE) / \text{Total technology expenditure}$.

For balanced search (BS), this paper refers to the studies of Cao et al. (2009) and Wong et al. (2013) to calculate the balance dimension; using the absolute value of search breadth and depth, the smaller the result, the more balanced it is. Its calculation formula is $BS = |Sb - Sd|$.

3.2.3. Moderating variables

There are two methods to measure absorptive capacity in existing studies, which mainly design the scale for questionnaire survey and use proxy variables. In this paper, based on the availability and reliability of data and following Cohen and Levinthal (1990); Xie and Li (2018) and Volberda et al. (2010) absorptive capacity was measured by the ratio of R&D expenditure to sales revenue in the year.

3.2.4. Control variables

Based on the existing research results, this paper takes variables such as the average size of industry, R&D expenditure intensity, R&D manpower input intensity, market competition degree and export intensity that may affect the innovation quality results of high-tech manufacturing firms as control variables.

- (1) Average size of industry. In existing studies, scale has always been considered to have an impact on innovation, but the average size of the industry is different from that of a firm. Firms with a large industry average size have more advantages in terms of manpower, capital and risk resistance and thus have more convenient conditions for R&D activities (Laursen and Salter, 2006). Segarra-Oña et al. (2011) found a positive relationship between average size of industry and innovation performance of a firm through empirical analysis. Therefore, this paper takes the average size of industry as the control variable and measures the ratio of the total number of employees in different industries to the number of firms in high-tech manufacturing.
- (2) Intensity of R&D expenditure investment. R&D expenditure is the main source of innovation funds, which provide the basis and possibility for technological innovation and directly affect the capability and performance of technological innovation. Existing studies show that investment in R&D funds is conducive to the creation of high value-added output, which is significantly positively with innovation performance (Wang et al., 2010). Therefore, according to existing studies, this paper takes the intensity of R&D expenditure investment as the control variable and measures the proportion of R&D expenditure in the main business income of the year.
- (3) Intensity of R&D human capital investment. In addition to R&D expenditure investment, R&D investment also includes R&D human capital investment. R&D human capital effect will directly affect innovation performance. Existing studies show that R&D personnel investment have significantly positively impact on innovation performance (Artz et al., 2010). Therefore, based on existing studies, this paper takes the intensity of R&D human capital investment as the control variable and chooses the proportion of total time equivalent of R&D personnel and total number of employees to represent that intensity.
- (4) Degree of market competition. The degree of cooperation and competition within the industry is closely related to the number of firms. The increase in the number of firms within the industry and the improvement of market competition will strengthen the awareness of crisis, promote the diffusion of knowledge and technology spillover and have an important impact on innovation performance (Guan and Chen, 2010). Therefore, this paper takes the degree of market competition within the industry as a control variable and uses the number of high-tech manufacturing firms to reflect the degree of market competition.

Table 1
Variables and measure.

Variable	Variable name	Variable measure
Dependent variable	Innovation quality (Iq)	Granted number of invention patents/number of patent applications
	Technology search breadth (Sb)	Technology introduction and purchase domestic technology expenditure / total technology expenditure
Independent variables	Technology search depth (Sd)	Digestion and absorption, technical transformation expense / total technology expenditure
	Balanced search (Bs)	Absolute value of the difference between search breadth and search depth
Moderating variable	Absorptive capacity (Ac)	R&D expenditure / annual sales revenue
	Average size of industry (Size)	Total number of employees / number of firms
	Intensity of R&D expenditure investment (Rde)	R&D expenditure / main business income
Control variables	Intensity of R&D human capital investment (Rdh)	R&D personnel full-time equivalent / total number of employees
	Degree of market competition (Mc)	Number of firms in high-tech manufacturing industry
	Intensity of export (Exp)	New product export sales revenue / new product sales revenue

- (5) Intensity of export. The export of high-tech manufacturing products is conducive to promoting the transfer of knowledge at home and abroad, and the differences in the export intensity of new products lead to different profits for firms. Rich profit returns will feed the innovation investment of firms, thus promoting improvement in innovation level and quality (Rogers, 2004). Therefore, according to the general practice in academia, this paper takes the intensity of export as a control variable and measures the proportion of export sales revenue of new products in the sales revenue of new products.

The definition of each variable in this paper is shown in Table 1.

3.3. Model

In this study, we use the panel data model to analyze the relationships between technology search, absorptive capacity, and innovation quality in high-tech manufacturing firms. We treat innovation quality as the dependent variable, technology search as the independent variable, and firm absorptive capacity as the moderating variable. To test the proposed hypotheses H₁–H₃, we establish the following regression models:

$$Iq_{i,t} = \alpha_0 + \alpha_1 Sb_{i,t-1} + \alpha_2 Sb_{i,t-1}^2 + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{1,i,t} + \varepsilon_{1,i,t} \quad (1)$$

$$Iq_{i,t} = \beta_0 + \beta_1 Sd_{i,t-1} + \beta_2 Sd_{i,t-1}^2 + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{2,i,t} + \varepsilon_{2,i,t} \quad (2)$$

$$Iq_{i,t} = \chi_0 + \chi_1 Bs_{i,t-1} + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{3,i,t} + \varepsilon_{3,i,t} \quad (3)$$

In the regression, all the independent variables and control variables are lagged by one period, mainly to consider the lag of the influence of the variables and exclude the interference of two-way causality between the variables. We test the inverted U-shaped relationships between the breadth and depth of technology search, and the innovation quality of firms. According to Haans et al. (2016), to establish an inverted U-shaped relationship, the following conditions must be met: First, the coefficients α_1 and β_1 of the search dimensions are significantly positive, the coefficients α_2 and β_2 of the search dimensions squares are significantly negative, and the joint significance test with nonzero values must pass. Second, the slope of the inverted U-shaped curve is steep enough at both ends of the sample data range. In other words, when technology search breadth and depth are taken as the minimum values, $\alpha_1 + 2\alpha_2 \times Sb_{\min} > 0$ and $\beta_1 + 2\beta_2 \times Sd_{\min} > 0$, and they are significant, while when technology search breadth and depth are taken as the maximum values, $\alpha_1 + 2\alpha_2 \times Sb_{\max} < 0$ and $\beta_1 + 2\beta_2 \times Sd_{\max} < 0$, and they are significant. Third, the inflection point of the curve must be within the range of the sample data.

To test hypotheses H_{4a}–H_{4c}, on the basis of Eqs. (1)–(3), we add the interaction terms of absorptive capacity, search dimension, and search dimension square to establish Eqs. (4)–(6), respectively, as follows:

$$Iq_{i,t} = \lambda_0 + \lambda_1 Sb_{i,t-1} + \lambda_2 Sb_{i,t-1}^2 + \lambda_3 Sb_{i,t-1} \times Ac_{i,t-1} + \lambda_4 Sb_{i,t-1}^2 \times Ac_{i,t-1} + \lambda_5 Ac_{i,t-1} + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{4,i,t} + \varepsilon_{4,i,t} \quad (4)$$

$$Iq_{i,t} = \gamma_0 + \gamma_1 Sd_{i,t-1} + \gamma_2 Sd_{i,t-1}^2 + \gamma_3 Sd_{i,t-1} \times Ac_{i,t-1} + \gamma_4 Sd_{i,t-1}^2 \times Ac_{i,t-1} + \gamma_5 Ac_{i,t-1} + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{5,i,t} + \varepsilon_{5,i,t} \quad (5)$$

$$Iq_{i,t} = \eta_0 + \eta_1 Bs_{i,t-1} + \eta_2 Bs_{i,t-1} \times Ac_{i,t-1} + \eta_3 Ac_{i,t-1} + \sum_i^j \delta \cdot Controls_{i,t-1} + \mu_{6,i,t} + \varepsilon_{6,i,t} \quad (6)$$

In Eqs. (1)–(6), in addition to the variables defined and explained above, α , β , χ , λ , γ , and η in the model are the regression

Table 2
Descriptive statistics of variables.

Variable	Mean	P50	SD	Min	Max
Iq	0.517	0.499	0.157	0.205	0.869
Sb	0.449	0.047	1.838	0.002	18.32
Sd	0.519	0.170	1.102	0.001	10.49
Bs	0.691	0.150	1.934	0.000	18.17
Ac	0.104	0.081	0.102	0.013	0.845
Size	4.661	4.641	0.392	3.872	5.920
Rde	0.043	0.019	0.099	0.005	0.916
Rdh	3.382	3.114	0.931	1.505	5.717
Mc	6.651	6.760	1.450	3.135	9.641
Exp	0.366	0.325	0.293	0.003	1.357

coefficients of the model intercept terms, the explanatory variable, and the control variables; i represents a different high-tech manufacturing firm, t represents the year, and $\sum \delta \bullet$ Controls refers to the set of the other control variables not shown in the model; μ is the random disturbance term; and ε is the residual term.

4. Empirical analysis

4.1. Descriptive statistics and correlation analysis

Table 2 lists the means, medians, standard deviations, minima, maxima, and numbers of observation of relevant variables. In the sample data, the mean of innovation quality is 0.517, greater than the median 0.499, and the maximum and minimum of innovation quality are 0.869 and 0.205, respectively, indicating that the distribution of the sample is right biased. The maximum values (18.32, 10.49, 18.17) and minimum values (0.002, 0.001, 0.000) of technical search breadth, depth, and balance search are significantly different, and the mean values (0.449, 0.519, 0.691) are all greater than the median values (0.047, 0.170, 0.150), indicating that different high-tech manufacturing firms have large differences and uneven distributions. The maximum and minimum values of absorptive capacity are 0.013 and 0.845, respectively, and the mean value is 0.104, indicating that the overall absorptive capacity of high-tech manufacturing firms is low. Data of the other control variables show little difference.

Table 3 shows the correlation coefficient matrix between the variables. As shown in the table, there is a significantly positive relationship between technology search breadth (Sb), depth (Sd), and balanced search (Bs), and innovation quality (Iq). However, the relationships between the independent variables and dependent variables need to be further explored through regression analysis. There is no significant correlation between absorptive capacity (Ac) and innovation quality (Iq) from the data results. In addition, the correlation coefficients between the variables are mostly lower than 0.5. However, We noticed that some correlation coefficients are greater than 0.5. In order to avoid the problem of multicollinearity, this paper conducts a variance inflation factor test (VIF test). Table 4 is the VIF test result of variables. Table 4 shows that the maximum value of VIF test result is $5.98 < 10$, the mean of VIF test result is $2.67 < 5$, therefore, there is no serious collinearity problem among the variables.

4.2. Unit root test and stationarity analysis

Since panel data is composed of time series data and cross-section data, a unit root test should be conducted on the stationarity of the variable data before regression to prevent model pseudo regression. The common unit root test methods mainly include LLC, ADF, PP, HT, IPS, HADRI etc. Based on the applicability of different methods, we applied LLC, HT and IPS to test the data. Considering that the test results of different methods are not the same, we consider that as long as two or more methods are considered stable among the three methods, the data are considered stable. Table 5 reports the results.

4.3. Analysis of regression results

The sample data selected in this paper are panel data. Before conducting the regression analysis, it is necessary to determine whether the regression equation is a traditional OLS model, fixed-effect model or random-effect model. To this end, we performed the Hausman test and Breusch-Pagan test. The Hausman test results accept the null hypothesis ($p = 0.838$), while the Breusch-Pagan test results reject the null hypothesis ($p = 0.000$). Therefore, it is more appropriate to use the random-effect model.

4.3.1. The effect of technology search on innovation quality in high-tech manufacturing firms

Table 5 reports the regression results of hypotheses H_1 , H_2 and H_3 . Column (1) is the regression results containing only control variables. Except for the intensity of the R&D expenditure investment variable, all the variables have a significantly positive impact on innovation quality. The intensity of the R&D expenditure investment is positively with innovation quality, but not significantly. The reason may be that the high-quality innovation output of the firm is a long-term accumulation process, the R&D expenditure investment cannot have an immediate effect, and the innovation itself has a time lag effect. The variables of technology search breadth

Table 3
Correlation coefficient matrix of variables.

Variables	Iq	Sb	Sd	Bs	Ac	Size	Rde	Rdh	Mc	Exp
Iq	1.000									
Sb	0.146**	1.000								
Sd	0.204***	0.093	1.000							
Bs	0.145**	0.901***	0.427***	1.000						
Ac	-0.063	0.231***	0.067	0.169**	1.000					
Size	-0.460***	-0.237***	-0.220***	-0.306***	-0.092	1.000				
Rde	0.000	0.537***	0.120*	0.417***	0.797***	-0.074	1.000			
Rdh	0.450***	0.164**	0.102	0.229***	-0.330***	-0.773***	-0.242***	1.000		
Mc	-0.058	-0.098	-0.311***	-0.130*	-0.446***	-0.319***	-0.389***	0.479***	1.000	
Exp	0.085	-0.050	-0.101	-0.074	-0.423***	0.311***	-0.227***	-0.010	0.045	1.000

Note: *, **, and *** indicate that they pass the test at the levels of 10 %, 5 %, and 1 %, respectively.

Table 4
Variable VIF test.

Variables	VIF	1/VIF	Variables	VIF	1/VIF
Sb	2.17	0.46	Rdh	2.03	0.49
Sd	5.98	0.17	Rde	2.46	0.41
Bs	4.31	0.23	Mc	1.14	0.88
Ac	1.53	0.65	Exp	2.08	0.48
Size	2.36	0.42	Mean	2.67	–

Table 5
Variable stationarity test.

Variables	LLC	HT	IPS	Test result
Iq	–23.99***	–0.174***	–1.566*	Stability
Sb	–43.03***	–0.292***	–0.517	Stability
Sd	–23.32***	–0.396***	–1.08	Stability
Bs	–33.68***	–0.332***	–0.604	Stability
Ac	–13.32***	–0.228***	–2.182**	Stability
Size	–32.84***	0.208	–3.052***	Stability
Rde	–12.38***	–0.035***	–0.698	Stability
Rdh	–1.2e+02***	0.242***	–1.751**	Stability
Mc	–16.74***	0.031***	0.118	Stability
Exp	–6.898***	–0.177***	–3.501***	Stability

Note: *, **, and *** indicate that they pass the test at the levels of 10 %, 5 %, and 1 %, respectively.

and depth are added in column (2), and the results show that the regression coefficients of technology search depth and breadth are both positive and significant at the level of 1% ($\alpha_1 = 1.653$, $p < 0.01$; $\beta_1 = 1.209$, $p < 0.01$), indicating that with the increase of technology search depth and breadth of firms, the innovation quality also improves. Column (3) is the regression after adding the search breadth and the quadratic term of search breadth. The results show that the search breadth coefficient is 1.194, which is significant at the 1% level and the search breadth quadratic term coefficient is -0.049, which is significant at the 1% level. When technology search breadth takes the minimum value (0.002), the slope of the curve $\alpha_1 + 2\alpha_2 \times Sb_{\min} = 1.194$ is greater than 0; when technology search breadth takes the maximum value (18.32), the slope of the curve $\alpha_1 + 2\alpha_2 \times Sb_{\max} = -0.601$ is less than 0, while the slope of the sample boundary has the opposite sign. The inflection point of the curve is 12.18, which is within the sample range. With increasing breadth of firm's technology search, innovation quality shows a trend of first increasing and then decreasing. Hypothesis H₁ stating that there is an inverted U-shaped relationship between technology search breadth and innovation quality is verified. Column

Table 6
Relationship between technology search and innovation quality.

Variables	(1)	(2)	(3)	(4)
Sb		1.653*** (0.389)	1.194*** (0.426)	
Sb ²			–0.049** (0.021)	
Sd		1.209*** (0.294)		0.611* (0.361)
Sd ²				–0.106*** (0.041)
Bs		–1.673*** (0.364)	–0.423** (0.201)	–0.256** (0.120)
Ac		1.204 (3.892)	–0.561 (4.013)	–5.890 (3.810)
Size	3.430*** (1.072)	3.441*** (1.081)	3.588*** (1.103)	3.950*** (1.119)
Rde	1.348 (2.287)	6.452* (3.569)	5.864 (4.055)	–3.778 (3.043)
Rdh	5.778* (2.963)	5.134* (3.040)	5.979* (3.103)	6.261** (3.116)
Mc	23.161*** (7.589)	19.655** (8.273)	25.953*** (8.108)	28.465*** (8.484)
Exp	1.891*** (0.686)	1.685** (0.682)	1.620** (0.703)	1.996*** (0.707)
_cons	14.591*** (4.832)	15.608*** (4.810)	14.623*** (4.854)	15.890*** (4.924)
R ² _O	0.361	0.414	0.402	0.399

Standard errors are in parentheses, *, **, and *** indicate that they pass the test at the levels of 10 %, 5 %, and 1 %, respectively.

(4) is the regression results after adding search depth and the quadratic term of search depth. The results show that the coefficient of technical search depth is 0.611, which is only significant at the level of 10 %. The coefficient of the search depth quadratic term is -0.106, which is significant at the 1% level. When technology search depth takes the minimum value (0.001), the slope of the curve $\alpha_1 + 2\alpha_2 \times Sd_{\min} = 0.611$ is greater than 0; when technology search depth takes the maximum value (10.49), the slope of the curve $\alpha_1 + 2\alpha_2 \times Sd_{\max} = -1.613$ is less than 0, while the slope of the sample boundary has the opposite sign. The inflection point of the curve is 2.882, which is within the sample range. Therefore, with increasing depth of firm technology search, innovation quality shows a trend of first increasing and then decreasing. Hypothesis H₂, stating that there is an inverted U-shaped relationship between technology search depth and innovation quality is verified. From columns (2) to (4), the sign of the balanced search and innovation quality relationship coefficient is negative and is significant at the 1% and 5% levels, respectively. This result does not mean that balanced search is negatively correlated with quality of innovation. The main reason for the negative sign is that balanced search is measured by the absolute value of the difference in search breadth and search depth. The greater the difference is, the worse is the balance and the lower is the innovation quality. Therefore, when firms conduct technology search, the balance of search depth and breadth dimension is better controlled, and the quality of innovation is relatively high. Hypothesis H₃ stating that balanced search is positively correlated with innovation quality is verified.

In addition, columns (2) to (4) in Table 6 show that the relationship between absorptive capacity and innovation quality is unstable and not significant, indicating that absorptive capacity has no direct impact on innovation quality. Therefore, we next analyze the moderating effects of absorptive capacity on the relationships between the variables.

4.3.2. The moderating effect of absorptive capacity

After introducing the interaction terms of technology search breadth, depth, and their square term with absorptive capacity into the equation in models (5) and (6), the inflection point of the curve is as follows:

$$Sb^* = \frac{-\lambda_1 - \lambda_3 \times Ac}{2\lambda_2 + 2\lambda_4 \times Ac}, \quad Sd^* = \frac{-\gamma_1 - \gamma_3 \times Ac}{2\gamma_2 + 2\gamma_4 \times Ac} \quad (7)$$

Obviously, the position of the inflection point changes with Ac. Calculate the partial derivative of Eq. (7):

Table 7
Moderating effect of absorptive capacity.

Variables	(5)	(6)	(7)
Sb	1.180* (0.653)		
Sb ²	-0.160*** (0.057)		
Sb × Ac	0.451*** (0.120)		
Sb ² × Ac	-0.056*** (0.016)		
Sd		0.943*** (0.936)	
Sd ²		-0.540*** (0.114)	
Sd × Ac		0.510*** (0.145)	
Sd ² × Ac		-0.091*** (0.021)	
Bs			-0.572** (0.270)
Bs × Ac			3.786* (2.085)
Ac	0.062** (0.093)	0.086** (0.094)	0.185** (0.093)
Size	4.831*** (1.276)	4.443*** (1.350)	5.009*** (1.304)
Rde	-0.724 (3.793)	-1.183 (2.313)	7.667 (4.802)
Rdh	1.742 (3.298)	4.619 (3.413)	3.678 (3.376)
Mc	22.717*** (8.001)	28.913*** (8.383)	28.621*** (8.079)
Exp	1.215* (0.709)	1.367* (0.700)	1.525** (0.719)
_cons	18.428*** (5.080)	15.949*** (5.253)	17.668*** (5.111)
R ² _O	0.410	0.426	0.391

Standard errors are in parentheses; *, **, and *** indicate that they pass the test at the levels of 10 %, 5 %, and 1 %, respectively.

$$\frac{\partial Sb^*}{\partial Ac} = \frac{\lambda_1\lambda_4 - \lambda_2\lambda_3}{2(\lambda_2 + \lambda_4 \times Ac)^2}, \quad \frac{\partial Sd^*}{\partial Ac} = \frac{\gamma_1\gamma_4 - \gamma_2\gamma_3}{2(\gamma_2 + \gamma_4 \times Ac)^2} \quad (8)$$

When Eq. (8) is not equal to 0, the inflection point of the curve moves because $2(\lambda_2 + \lambda_4 \times Ac)^2$ and $2(\gamma_2 + \gamma_4 \times Ac)^2$ are always greater than zero. Therefore, when $\lambda_1\lambda_4 - \lambda_2\lambda_3 < 0$ and $\gamma_1\gamma_4 - \gamma_2\gamma_3 < 0$, the inflection point shifts to the left; when $\lambda_1\lambda_4 - \lambda_2\lambda_3 > 0$ and $\gamma_1\gamma_4 - \gamma_2\gamma_3 > 0$, the inflection point shifts to the right.

Hypotheses H_{4a} and H_{4b} indicate that absorptive capacity plays a moderating role in the inverted U-shaped relationships between technology search breadth and depth, and firms' innovation quality. The higher the level of absorptive capacity is, the greater are the positive influences of technology search breadth and depth on the firm's innovation quality. Compared with a low absorptive capacity firm, a high absorptive capacity firm's curve inflection point moves to the right, with the expectation that $\lambda_1\lambda_4 - \lambda_2\lambda_3 > 0$ and $\gamma_1\gamma_4 - \gamma_2\gamma_3 > 0$. Hypothesis H_{4c} indicates that absorptive capacity positively moderates the relationship between balanced search and innovation quality. The stronger it is, the greater is the positive impact of balanced search on innovation quality. Incidentally, the slope of the curve of the relationship between balanced search and innovation quality will change.

As shown by the results of each model in Table 7, the coefficients of technology search breadth and depth are positive, and the coefficients of the quadratic terms are negative, and they are all significant. Therefore, after including the interaction item, the inverted U-shaped relationship between technology search and innovation quality is still valid.

Columns (5) and (6) in Table 7 are the model regression results after adding the interaction terms of technology search breadth, depth, and their quadratic terms with absorptive capacity, respectively. The coefficients of the interaction terms of technical search breadth and depth with absorptive capacity are significantly positive ($\lambda = 0.451, p < 0.01$; $\gamma = 0.943, p < 0.01$), and the coefficients of the interaction terms of technology search breadth quadratic terms and depth quadratic terms with absorptive capacity are significantly negative ($\lambda = -0.056, p < 0.01$; $\gamma = -0.091, p < 0.01$). The results show that absorptive capacity strengthens the positive effects of technology search breadth and depth on innovation quality. Next, we determine whether the inflection point has changed by noting that $\lambda_1\lambda_4 - \lambda_2\lambda_3 = 0.006 > 0$ and $\gamma_1\gamma_4 - \gamma_2\gamma_3 = 0.189 > 0$. Therefore, the values of $\frac{\partial Sb^*}{\partial Ac}$ and $\frac{\partial Sd^*}{\partial Ac}$ are both greater than 0, and the inflection point of the curve moves to the right. We draw a moderating effect diagram of absorptive capacity according to the results in Table 7.

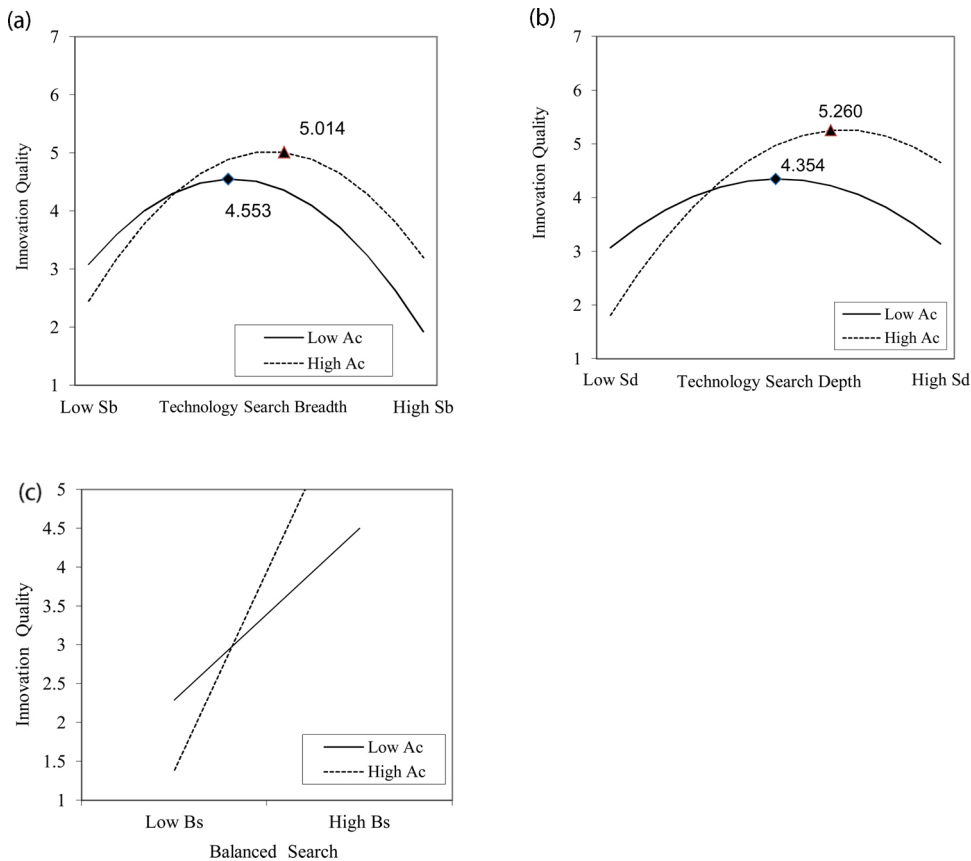


Fig. 3. (a) Moderating effect of absorptive capacity on the relationship between technology search breadth and innovation quality. (b) Moderating effect of absorptive capacity on the relationship between technology search depth and innovation quality. (c) Moderating effect of absorptive capacity on the relationship between balanced search and innovation quality.

Fig. 3(a) shows changes in the relationship curve between technology search breadth and innovation quality of high-tech manufacturing firms before and after considering the moderation of absorptive capacity. (1) Before the inflection point, the curve at a low absorptive capacity level is relatively flat, while the curve of at a high absorptive capacity level is steep, indicating that high-tech manufacturing firms with a high absorptive capacity level improve their innovation quality faster with increasing search breadth. (2) The position of the inflection point changes; compared with the inflection point of the low absorptive capacity curve (4.553), the inflection point of the high absorptive capacity curve (5.014) moves to the upper right, and the vertex of innovation quality is also higher, indicating that the higher the absorptive capacity level of high-tech manufacturing firms is, the more conducive is the technology search breadth to improvement of innovation quality. (3) After the inflection point, the curves of innovation quality begin to decline. Although the flattening degrees with different level of absorptive capacity curves are similar, but the curve with high absorptive capacity is always above the one with low absorptive capacity, indicating that even with the same search breadth, the innovation quality of firms with high absorptive capacity is higher than that of firms with low absorptive capacity.

It cannot be concluded that absorptive capacity per se positively moderates the relationship between technology search breadth and innovation quality of high-tech manufacturing firms because the moderating effect is related to the extent of technology search breadth. At a lower level of technology search breadth, absorptive capacity negatively moderates the relationship between them. When the extent of technology search breadth is high, absorptive capacity positively moderates the relationship between them. Therefore, the results of this paper only partially support the hypothesis H_{4a}.

Fig. 3(a) shows changes in the relationship curve between technology search depth and innovation quality of high-tech manufacturing firms before and after considering the moderation of absorptive capacity. (1) Before the inflection point, the curve at a low absorptive capacity level is relatively flat, while the curve of at a high absorptive capacity level is steep, indicating that high-tech manufacturing firms with a high absorptive capacity level improve their innovation quality faster with increasing search depth. (2) The position of the inflection point changes; compared with the inflection point of the low absorptive capacity curve (4.354), the inflection point of the high absorptive capacity curve (5.260) moves to the upper right, and the vertex of innovation quality is also higher, indicating that the higher the absorptive capacity level of high-tech manufacturing firms is, the more conducive is the technology search depth to improvement of innovation quality. (3) After the inflection point, the curves of innovation quality begin to decline, but the curve with high absorptive capacity is always above the one with low absorptive capacity, indicating that innovation quality declines with the improvement of technology search depth, regardless of the level of absorptive capacity; when the extent of technology search depth is high, the innovation quality of High-tech manufacturing firms is higher in the context of high absorptive

Table 8
Robustness test results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Sb	2.347*** (0.660)	−0.476* (0.274)			0.279*** (0.087)			0.341** (0.147)	
Sb ²					−0.038*** (0.015)			−0.048** (0.020)	
Sd			0.611* (0.361)	−2.427** (0.395)		0.020** (0.009)			0.125* (0.070)
Sd ²						−0.002** (0.001)			−0.055*** (0.018)
Sb × Ac								2.118* (1.258)	
Sb ² × Ac								−0.162 (0.126)	
Sd × Ac									1.281* (0.623)
Sd ² × Ac									−0.500*** (0.160)
Bs							0.043*** (0.015)		
Ac	0.190 (3.994)	−6.705 (10.184)	−5.890 (3.810)	2.149 (6.367)			4.485*** (0.551)	9.632*** (1.234)	4.133*** (0.693)
Size	3.420*** (1.090)	−2.455 (9.512)	3.950*** (1.119)	4.366*** (1.291)	1.111*** (0.351)	0.020 (0.030)	0.027 (0.174)	0.818** (0.346)	0.124 (0.179)
Rde	9.445** (4.294)	−1.417 (11.864)	−3.778 (3.043)	−0.533 (10.309)	5.519*** (1.116)	0.374*** (0.073)	2.418*** (0.416)	15.998*** (3.285)	2.144*** (0.539)
Rdh	3.945 (3.101)	34.153*** (12.593)	6.261** (3.116)	6.385 (4.192)	1.485*** (0.164)	0.087*** (0.014)	0.534*** (0.083)	1.251*** (0.170)	0.624*** (0.082)
Mc	20.510** (8.146)	55.109 (54.693)	28.465*** (8.484)	48.348*** (11.737)	4.414* (2.344)	0.019** (0.008)	2.296* (1.382)	5.095** (2.194)	3.069** (1.341)
Exp	1.368** (0.695)	4.154 (3.822)	1.996*** (0.707)	2.558*** (0.742)	−0.545** (0.220)	0.015 (0.017)	−0.139 (0.100)	−0.641*** (0.209)	−0.093 (0.101)
_cons	14.581*** (4.871)	−5.424* (25.316)	15.890*** (4.924)	9.116* (6.304)	0.307*** (1.533)	−0.203*** (0.126)	1.079* (0.857)	0.341** (1.424)	0.670** (0.822)
R ² _O	0.401	0.710	0.400	0.435	0.558	0.498	0.520	0.621	0.573

Standard errors are in parentheses; *, **, and *** indicate that they pass the test at the levels of 10 %, 5 % and 1 %, respectively.

capacity.

It cannot be concluded that absorptive capacity per se positively moderates the relationship between technology search depth and innovation quality of high-tech manufacturing firms because the moderating effect is related to the extent of technology search depth. At a lower level of technology search depth, absorptive capacity negatively moderates the relationship between them. When the extent of technology search depth is high, absorptive capacity positively moderates the relationship between them. Therefore, hypotheses H_{4b} are partially supported.

Column (7) is the model after adding the interaction term between balanced search and absorptive capacity. The results show that the interaction coefficient is positively significant ($\eta = 3.786, p < 0.1$). Thus, hypothesis H_{4c} is supported. Fig. 3(c) shows that the curve at a high absorptive capacity level is steeper, indicating that high-tech manufacturing firms at a high absorptive capacity level improve their innovation quality faster with the enhancement of balanced search.

4.4. Robustness tests

The relationships between the search breadth and depth, and innovation quality of high-tech manufacturing firms and the moderating effect of absorptive capacity are empirically analyzed. To ensure the robustness of the research results, we conducted robustness tests on the following two aspects, and report the results in Table 8.

- (1) We measure the independent variable groups separately. Following Haans et al. (2016) for the U-shaped and inverted U-shaped test recommendations, we divided the samples of technology search breadth and depth into two groups based on the inflection point of the curve for regression, respectively. If the data on the left are significantly positively correlated with innovation quality and the data on the right are significantly negatively correlated, the inverted U-shaped relationship is valid. In Table 8, columns (1) and (3) are the regression results of datasets with technology search breadth and depth less than the inflection point, which are positively correlated with innovation quality ($\alpha = 2.347, p < 0.01$; $\beta = 0.611, p < 0.1$). Columns (2) and (4) are the regression results of datasets with technology search breadth and depth greater than the inflection point, which are negatively correlated with innovation quality ($\alpha = -0.476, p < 0.1$; $\beta = -2.427, p < 0.05$). Thus, with increasing technology search breadth and depth, firms' innovation quality shows a trend of first increasing and then decreasing. The above results further support hypotheses H₁ and H₂ that technology search breadth and depth each have an inverted U-shaped relationship with innovation quality.
- (2) We change the measurement of the dependent variable. Innovation quality is the sum of innovation performance in various fields, and the relationship between innovation quality and innovation performance is qualitative and quantitative. Therefore, we take innovation performance as an alternative indicator of innovation quality to re-conduct the regression estimation. According to Chen and Miller (2007), innovation performance is measured by the sales of new products or services that are produced by innovation as a percentage of total revenue. We report the regression results in columns (5) to (9) of Table 8. The main effect test results in columns (5) and (6) show that the coefficients of technology search breadth and depth are significantly positive ($\alpha = 0.279, p < 0.01$; $\beta = 0.020, p < 0.05$), and the coefficients of technology search breadth and depth square terms are significantly negative ($\alpha = -0.038, p < 0.01$; $\beta = -0.002, p < 0.05$). The slopes of the sample range on the left are $\alpha_1 + 2\alpha_2 \times Sb_{\min} = 0.341$ and $\beta_1 + 2\beta_2 \times Sd_{\min} = 0.125$, both of which are greater than 0. The slopes on the right are $\alpha_1 + 2\alpha_2 \times Sb_{\max} = -1.418$ and $\beta_1 + 2\beta_2 \times Sd_{\max} = -1.029$, both of which are less than 0; the inflection points of the curve are 3.552 and 1.136, respectively, both of which are within the range of the sample values. There are still significant inverted U-shaped relationships between technology search breadth and depth, and innovation quality. The results in column (7) show that there is a significantly positive relationship between balanced search and innovation quality.

Columns (8) and (9) reflect the moderating effect of absorptive capacity. The results show that the coefficients of the interaction terms between technology search breadth and depth, and absorptive capacity are significantly positive ($\lambda = 2.118, p < 0.1$; $\gamma = 1.081, p < 0.1$), the coefficient of search breadth square term and absorptive capacity interaction term is negative but not significant ($\lambda = -0.162, p > 0.1$), and the coefficient of the technology search depth squared term and absorptive capacity interaction term is significantly negative ($\gamma = -0.500, p < 0.01$). Similarly, whether the inflection point of the curve changes is determined based on the results of Table 8, i.e., $\lambda_1\lambda_4 - \lambda_2\lambda_3 = 0.157$ and $\gamma_1\gamma_4 - \gamma_2\gamma_3 = 0.008$. Therefore, the values of $\frac{\partial Sb^*}{\partial Ac}$ and $\frac{\partial Sd^*}{\partial Ac}$ are both greater than 0, and the inflection point of the curve moves to the right, consistent with the previous results, showing that the research results have strong robustness.

5. Discussion and conclusion

5.1. Conclusion

Concerning the influence of technology search behavior on firm innovation, some scholars believe that there are positive effects (Leiponen and Helfat, 2010; Phelps, 2010), and some results show negative effects (Knudsen and Mortensen, 2011; Swift, 2016). Some scholars also propose that there is a nonlinear relationship between technology search and innovation (Laursen and Salter, 2006; Wu, 2013). This paper divides technology search into three dimensions: search breadth, depth and balanced search. Combined with the above viewpoints and exploring the benefits and costs of different search dimensions, according to the dynamic change of the

relationship, we propose the hypotheses of an inverted U-shaped relationship between the technology search breadth, depth and innovation quality of high-tech manufacturing firms as well as the hypothesis of a linear relationship between balanced search and innovation quality of high-tech manufacturing firms.

This paper takes the data of China's high-tech manufacturing industry from 2009 to 2017 as the sample, and the research conclusions are as follows: First, The relationship between technology search width and innovation quality, technology search depth and innovation quality is inverted U-shape. After introducing absorptive capacity as a moderating variable, we find that the inflection point shifts to the upper right, which confirms that the higher the absorptive capacity of high-tech manufacturing firms, the more technical search breadth or technical search depth is conducive to the improvement of firm innovation quality. The moderating effect of absorptive capacity is closely related to the level of breadth or depth of technology search in high-tech manufacturing firms. At a lower level of technology search breadth or technology search depth, absorptive capacity negatively moderates the relationship between them. When the extent of technology search breadth or technology search depth is high, absorptive capacity positively moderates the relationship between them. Second, the balance search is positively with the innovation quality of high-tech manufacturing firms, and high absorptive capacity increases the slope of the innovation quality curve and makes it steeper. In other words, the higher a firm's absorptive capacity level is, the faster the balanced search will make the innovation quality increase

5.2. Theoretical contributions and management implications

We make three theoretical contributions in this study as follows: First, it further enriches the research of innovation management. Previous researches mainly focus on firm innovation activities from the perspective of quantity (Fleming and Sorenson, 2004), while few focus on innovation quality (Duan et al., 2021; Yu et al., 2019). Based on the knowledge base view, we construct the relationship model of technology search width, technology search depth, balance search and innovation quality, and verify the inverted U-shaped relationship between width and depth of technology search and innovation quality, and the linear relationship between balance search and innovation quality. Our conclusion is consistent with previous research (Laursen and Salter, 2006; Wu, 2013), which further supports the demonstration of the relationship between technology search and innovation quality.

Second, in terms of empirical methods, this paper provides more rigorous demonstration operation. For the demonstration of the inverted U-shaped relationship, previous studies mainly focused on the quadratic coefficient and significance of the principal variable. In addition, this paper verified the change in the slope of the sample boundary with the maximum and minimum values, and proved whether the change in the inflection point falls within the range of the variable, so as to strengthen the test of the nonlinear relationship. For the demonstration of the moderating variable, we not only pay attention to the coefficient and significance of the interaction term between the moderating variable and the core variables, but also determine whether the inflection point has changed to verify the moderating effect of absorptive capacity.

Third, it is found that absorptive capacity does not positively moderate the relationship between technology search and innovation quality, but is closely related to firms' technology search behavior. Absorptive capacity plays a different moderating role for different degree of technology searching. This is very different from the previous conclusion that absorptive capacity simply takes a positive or negative moderating effect, and it also provides valuable ideas and new ideas for further research.

Our research findings are helpful to high-tech manufacturing firm managers in that the results provide guidance and serve as a reference for them to make innovative strategic decisions for technology search.

First, the high-tech manufacturing firms should focus more on innovation quality in the process of innovation. However, to improve innovation quality, firms need to search for external technical knowledge. We find that there are inverted U-shaped relationships between technology search breadth and depth, and innovation quality, i.e., technology search improves innovation quality. There are optimal technical search breadth and depth, and firms should make the best effort not to deviate from such optimal critical points.

Second, there is a significantly positive correlation between balanced search and innovation quality, which means that high-tech manufacturing firms should focus on not only the optimal critical points of search breadth and depth, but also the balance point between search breadth and search depth. On the one hand, it is necessary that firms expand their search breadth to obtain a wider scope of heterogeneous knowledge elements and to understand the development trend of markets and technologies. On the other hand, they should enhance the depth of search for familiar knowledge, isolate tacit knowledge, create technical knowledge that is difficult to imitate, create technical barriers, and maintain a continuous competitive advantage.

Finally, while focusing on improvement of innovation quality, high-tech manufacturing firms should also focus on enhancing their own absorptive ability, establishing a knowledge management system, developing a good organizational learning mechanism, and cultivating and improving the ability of employees to acquire and apply knowledge.

5.3. Limitations and directions for future research

First, the sample data of this study are from high-tech manufacturing industry with strong innovation ability, and the relationship between technology searching behavior and innovation quality of all firms in the same category or the same industry is discussed. To some extent, this may ignore the heterogeneity of firms (Caner et al., 2017). In future research, we can obtain specific corporate data through field research, and study the relationship between different firm's technology search behaviors and innovation quality. Second, this paper only analyzes the direct relationship between technology search and innovation quality, so the issue as to whether there is a variable that mediates the relationship is worth further study. Thirdly, we refer to existing researches and consistent with scholars such as Duan et al. (2021) and Yu et al. (2019) to use patent ratio to measure firm innovation quality. There is no denying that such a measurement method is controversial, and a more scientific and accurate measurement method can be found in future research.

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